

Ali Haggag

Full-Stack Engineer · AI Agent Systems · Real-Time Systems · Compiler Design

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Last updated: August 2026 · This document is a living technical profile — send it to any AI or team to get full context on Ali's stack, projects, and battle scars.

// WHO IS ALI

Ali Haggag is a Computer Science & AI student (South Valley National University, 2023–2027) and a self-driven full-stack engineer who has shipped production systems across real-time multiplayer games, social super-apps, developer ecosystems, secure API backends, and autonomous AI agents. He specialises in system design, performance optimisation, compiler construction, WebSocket/WebRTC synchronisation layers, and LLM-orchestrated agentic systems with worker-thread browser automation. He communicates in Arabic (native) and English (professional working proficiency).

// ACTIVE PROJECTS (as of August 2026)

Scout [AI Autonomous Job-Application Agent — graduation project]

pnpm/Turborepo monorepo · NestJS 11 · Next.js 16 · Tauri v2 (Rust) desktop wrapper · PostgreSQL + Prisma · Redis + BullMQ · Socket.io · Playwright · Zod · Groq Llama-3.3-70B (BYOK, OpenAI-compatible fallback)

- Cognitive Brain Loop (Observe → Reason → Plan → Act → Verify) running on the main thread, with Playwright browser automation isolated inside node:worker_threads for crash-safe, concurrent execution.
- Hard ethical guardrails baked into the architecture: no server-side mass scraping, no residential proxy/IP rotation, no anti-bot evasion infrastructure, no third-party credential storage, and a mandatory Final Click Rule requiring explicit human approval before any submission.
- AI Application Intelligence pipeline: two-axis question classifier, deterministic-first answers (DB lookups, date arithmetic, option fuzzy-matching) with zero LLM cost, and a "Never Lie" claims verifier that fails and retries on unverifiable or fabricated claims.
- Career Memory & Strategy Recommendation engine — tracks outcome success rates across 5 dimensions and proposes preference adjustments through a human-approved recommendation flow (agents can never write search preferences directly).
- Multi-platform scraper adapters (Wellfound, Wuzzuf, Greenhouse, LinkedIn) with human-like pacing, anti-detection hardening, and real-DOM screening-question extraction.
- Live Observability Control Room: 1-2 FPS browser frame streaming, a real-time AI "thought stream," and a GitHub-Actions-style cognitive step timeline, all relayed over Socket.io.
- BYOK AI provider layer with automatic HTTP 429 fallback, AES-256-GCM field-level encryption for CVs/PII, and Pino structured logging with credential redaction.
- 156+ automated tests (Vitest/Jest) covering session lifecycle, worker isolation, and the full cognitive cycle.

Logic Arena [logicarena.dev]

v3.6.5 · TypeScript monorepo (pnpm workspaces) · Next.js 16 client · NestJS 11 server · React Three Fiber · Web Audio API · Socket.io · Redis · PostgreSQL + Prisma · Docker + Nginx · DigitalOcean · Cloudinary · Resend

- Custom DSL: AliScript v2.4 — full Lexer → AST parser → secure server-side evaluator. No eval(). Every script runs through a node-by-node interpreter under a deterministic 2,000 ops/tick quota (not wall-clock timing) — identical execution limits on every player's hardware, from a Raspberry Pi to a workstation.
- AliScript power layer: WHILE/IF/FOR, user-defined functions, Dictionary/state-machine objects, 8 GET_X() query functions, a Swarm Intelligence API (BROADCAST/RECEIVE with deep-copy sandbox isolation), and 6 tactical super powers — SHIELD, CLOAK, DASH, TELEPORT, MINE, TAUNT — each with real energy costs and cooldowns.
- Inline compile-time diagnostics: a worker-thread semantic analyzer flags logic contradictions, dead code, and unknown commands with Levenshtein-distance autocorrect — red wavy underlines and Tab-to-fix, live in the editor.
- AI Script Generator: natural-language (EN/AR, any language) → working AliScript via SSE-streamed AI completions, backed by a 22-recipe Strategy Cookbook across 3 difficulty tiers — lets non-programmers play.
- Mobile Block Editor: a full drag-and-drop visual programming system (@dnd-kit) that eliminates the on-screen keyboard entirely in landscape mode, with a recursive block-to-AliScript compiler.
- Game modes: Combat 1v1/FFA, Capture, Survival, Racing, King of the Hill, Practice vs AI (15 hand-scripted bot behaviors × 3 difficulty tiers with an algorithmic points-reward matrix), Tactical Mode (round-based combat with live mid-match script rewriting and a token-budget economy), Campaign (60 LeetCode-style algorithmic levels), and Tournaments (4/8-player brackets).
- Live-streamed 2D Campaign battles: the server runs a headless 60fps-physics/10Hz-logic simulation and streams frames straight to the client canvas at 20fps, with a hidden _SYS_* rail-system variable layer giving enemy AI pixel-precise scripted movement that's completely invisible at the language level.
- Cinematic 3D arenas: Cyber City / Glacial Tundra / Volcanic Core / Black Hole environments with theme-specific procedural assets, an infinite parallax starfield, and a fully synthesized 8-soundscape spatial audio engine built on pure Web Audio (zero MP3s, logarithmic 3D distance falloff).
- 20 TPS server physics engine decoupled from rendering via a sub-frame InterpolationBuffer — Three.js meshes are mutated directly

inside useFrame, delivering buttery-smooth 120 FPS movement independent of network tick rate or player ping. Delta-state diffing cuts WebSocket payload size ~80%.

- Economy: Black Market with COMMON→ELITE rarity tiers, chassis unlocks, paints, tracer rounds. Full PWA with fullscreen mobile install, offline pages, and native-feeling pull-to-refresh.
- 4-layer security: HttpOnly JWT cookies + token versioning, Redis rate-limiting, AST sandbox (hardened against prototype pollution and ReDoS), Prisma mass-assignment guards, DOMPurify XSS sanitization.
- Achievements & Badges: a 4-tier progression system with self-healing award checks that run across match, campaign, and cache-miss events.
- Admin Command Center: a full analytics dashboard (KPI cards, Recharts area/bar/donut charts, DB/Redis health gauges) behind its own dedicated rate-limit tier so monitoring never throttles itself.
- Campaign: server-authoritative CampaignSession pause/resume via incremental timestamp-shifting — every Date.now()-based entity (wall-stun timers, mine arming, shield windows) is re-anchored by the exact paused duration — plus a replay scrubber with adaptive canvas overlay.
- Cinematic camera system: dual-mode toggling (Free Orbit vs. Locked Robot Eye) and a dedicated first-person mode, plus a rate-independent lerp-based follow controller and a majestic capped gravitational camera pull near the Black Hole environment — all running at a jitter-free 60fps.
- 30+ battle-tested production incidents resolved across the engine, renderer, and infrastructure layers (see Battle Scars section).

Portfolio [alihaggag.me]

Next.js · TypeScript · Framer Motion · CSS variables · GitHub GraphQL API

- Interactive Hero terminal (AliOS) with 10+ custom commands: whoami, skills, projects, neofetch, sudo hire-me.
- Sci-fi visual identity: animated gradient mesh, twinkling stars via useMemo, glassmorphism, CRT scanlines.
- Battle Scars section documents real engineering incidents in SYMPTOM / CAUSE / FIX format.
- GitHub Activity card using REST + GraphQL with explicit .ok and gql.errors checks to prevent silent zero-data failure.
- app/error.tsx global boundary, BattleScarsSkeleton for CLS prevention, Chatbot error state surfacing.
- Strict rules enforced: CSS variables only (zero hardcoded hex), 44px min touch targets, Framer Motion disabled on mobile via useIsMobile, type='button' on all non-nav buttons.

Arshivi [personal archive app — in progress]

Vanilla HTML/JS/CSS · Supabase (free tier backend) · Cloudflare Pages · private GitHub repo

- Three-level architecture: Categories → Sections → Items.
- Converting a static HTML file into a full-stack app. Data migrated via Node.js script.
- Current work: custom dropdown + date picker, removing loading states between category navigation.

// PREVIOUS PROJECTS

Flurry v2.0 [flurry-app.vercel.app]

React · Socket.io · WebRTC · Google Gemini AI · MongoDB · PWA (Workbox) · i18next · Clerk · ImageKit

- WebRTC P2P audio/video calls via custom Socket.io signaling — <50ms latency.
- Offline-first PWA: IndexedDB action queue + Workbox service worker auto-sync.
- Gemini AI group chat summaries, real-time polls, CRUD messaging, voice notes.
- 4-state message delivery: Pending → Sent → Delivered → Read.
- Full EN/AR bilingual with dynamic RTL/LTR layout switching.
- Hard block system, private accounts, active-status control.

CS Arena [csarena.tech]

Next.js 16 · TypeScript · Sanity CMS · NextAuth · Sentry · Resend · Framer Motion · next-intl

- Developer project showcase platform with 3-level cascading classification (Domain → Specialization → Tech).
- Solved cascading filter race conditions using useTransition + URL-first state engine.
- Join-request collaboration system with Resend transactional emails.
- 3D Podium leaderboard UI, real-time GitHub stats, Sentry error tracking + session replay.
- PWA, auto-generated sitemap, bilingual AR/EN docs (MDX).

Blog Pro [blog-pro-platform.vercel.app]

React · Redux Toolkit · Express.js · MongoDB · Helmet · XSS-Clean · JWT · Cloudinary · Nodemailer

- 5-layer security pipeline: Helmet, XSS-Clean, HPP, Joi validation, Redis rate-limiting.
- RBAC admin dashboard with analytics, user/post/comment management.
- Nested comments, real-time likes, avatar uploads, follow/unfollow system.

Cybership Carrier API

TypeScript · Zod · DDD · Jest · Nock · Axios · OAuth 2.0

- Domain-Driven Design with an Anti-Corruption Layer (Zod) separating UPS carrier JSON from core domain.
- OAuth 2.0 singleton token client with 60-second expiry buffer — zero inflight request interruptions.
- Structured custom error classes: AuthenticationError, RateLimitError.

- Full test suite via Jest + Nock (no live credentials needed).

Realtime Chat Engine

React · Vite · Firebase (Firestore + Auth) · Cloudinary · Context API

- Serverless architecture — zero-maintenance scalability via Firebase BaaS.
- Instant message sync via Firebase real-time listeners (no page reloads).

// TECHNICAL ARSENAL

Frontend ● BATTLE-TESTED

Next.js 15/16 · React 19 · TypeScript · JavaScript ES6+ · Tailwind CSS · Redux Toolkit · Framer Motion · shadcn/ui · Magic UI · React Three Fiber / Three.js

Backend ● PRODUCTION READY

NestJS 11 · Node.js · Express.js · PostgreSQL · MongoDB · Redis · Firebase · Supabase · Prisma ORM

AI Agents & Automation ● BATTLE-TESTED

Groq (Llama 3.3 70B) · LLM Orchestration & Prompting (BYOK, multi-provider fallback) · Playwright · worker_threads isolation · BullMQ · Cognitive Agent Loop Design (Observe/Reason/Plan/Act/Verify) · Tauri v2 (Rust)

Real-Time & AI ● BATTLE-TESTED

Socket.io · WebRTC (SimplePeer) · Google Gemini API · PWA / Workbox / Service Workers

ORM & API Layer ● BATTLE-TESTED

Prisma · Zod · GraphQL · Redis (ioredis) · Inngest (durable queues) · BullMQ (Redis-backed queues)

Auth & Security ● PRODUCTION READY

JWT (HttpOnly cookies) · NextAuth v5 · Clerk · OAuth 2.0 (Google/GitHub) · Helmet · XSS-Clean · Rate Limiting · AES-256-GCM field-level encryption

DevOps & Infra ● PRODUCTION READY

Docker + Nginx · Docker Compose (multi-service) · DigitalOcean Droplets · Vercel · Cloudflare Pages · Sevala · pnpm Workspaces + Turborepo · Git / GitHub / GitLab

CMS & State ● PRODUCTION READY

Sanity CMS · Zustand · React Query / TanStack Query · i18next / next-intl

Email & Media ● PRODUCTION READY

Resend · Nodemailer · Mailtrap · Cloudinary · ImageKit CDN

CS & Compiler ● BATTLE-TESTED

AST Design & Compiler (AliScript) · Deterministic TLE quotas · A* Pathfinding (Weighted Grid) · C++ (competitive programming)

Monitoring & Logging ● PRODUCTION READY

Sentry (error tracking + session replay) · Pino (structured JSON logging, redaction) · Azure Monitor (R&D)

// BATTLE SCARS (real production incidents — SYMPTOM / CAUSE / FIX)

The Ghost Worker Uprising

[CRITICAL] Scout v0.24

Badges: worker_threads · Playwright · Session Lifecycle

SYMPTOM: Browser windows kept reopening and closing in a loop after "Stop Agent" was clicked, even though the UI showed IDLE.

CAUSE: Session.stop() flipped state to STOPPED but never terminated the background node:worker_threads; when the browser context closed, Playwright errors triggered RecoveryManager to relaunch, spawning zombie workers.

FIX: Added Executor.terminateAllWorkers() to force-kill active worker threads on stop/restart, plus a STOPPED guard check at the top of every brain cycle.

The Ghost Match Massacre

[CRITICAL] Logic Arena v2.5.0

Badges: WebSocket · NestJS · Memory Leak

SYMPTOM: Server kept executing MOVE_FAST for disconnected players, pinning CPU at 100%.

CAUSE: WebSocket dropped on disconnect but the MatchEngine physics loop kept running indefinitely, accumulating hundreds of dead matches.

FIX: Tied match lifecycle to connected client count — numClients === 0 calls match.stop() and destroys engine from RAM.

The Ghost Robot Infestation

[CRITICAL] Logic Arena v3.5.0

Badges: R3F · WebSocket · State Management

SYMPTOM: After rapid arena switching, robots from previous match appeared as invisible ghosts; on third switch nothing worked at all.

CAUSE: Module-level InterpolationBuffer singleton never cleared between matches; stale selectedRobotId/socketUserId refs silently rejected by server.

FIX: Added clear() on arena mount/unmount, reset all tracking refs in useGameState, replaced Link back buttons with router.push() emitting leaveMatch before navigation.

The Mobile Scroll Lag Catastrophe

[HIGH] Logic Arena v3.2.0

Badges: 3D Performance · CSS · Mobile

SYMPTOM: Scrolling the landing page and dashboard on mobile was severely stuttery, especially on mid-range Android devices.

CAUSE: A triple performance cascade: 3D background objects were recreated every frame (GPU memory churn), backdrop-blur forced a full-viewport compositor repaint on every scroll frame, and theme CSS transitions fired mid-scroll causing frame drops.

FIX: Wrapped 3D objects in useMemo (created once), removed backdrop-blur on mobile in favor of solid backgrounds (highest-impact fix), globally disabled CSS transitions during theme changes, and added an IntersectionObserver to pause 3D rendering when the canvas scrolls out of view.

The LAVA/ICE Damage Double-Stack

[HIGH] Logic Arena v3.2.0

Badges: Collision Physics · Game Balance

SYMPTOM: Robots standing in lava zones on LAVA/ICE themed maps took 15 HP/sec instead of 10 — instantly melting on contact.

CAUSE: Static CYBER-theme lava obstacles were leaking into LAVA/ICE maps and applying damage simultaneously with the dynamic LAVA_POOL obstacles; rotated obstacle hitboxes also used AABB collision, extending well past their visual boundary.

FIX: Removed static lava obstacles on LAVA/ICE themes to eliminate double-stacking, and upgraded TRAP/LAVA collision from AABB to properly rotation-aware Oriented Bounding Box (OBB) detection.

The VDOM Stuttering Catastrophe

[HIGH] Logic Arena v3.5.0

Badges: React · Three.js · Performance

SYMPTOM: Robot movement had a 'live stream buffering' quality — smooth, then stuttering, then smooth — especially on mid-range Android.

CAUSE: React's default shallow prop comparison detected changes on every 20Hz server tick for RobotModelInner and LaserModel, forcing full VDOM reconciliation and GC pauses from constant object churn.

FIX: Implemented custom React.memo equality functions that ignore rapid position/velocity/rotation prop mutations; Three.js meshes are now mutated directly inside useFrame, fully bypassing React's render lifecycle for movement.

The Spatial Audio Volume Crush

[HIGH] Logic Arena v3.5.0

Badges: Web Audio API · Signal Routing

SYMPTOM: After unifying all sound through one global AudioContext, environmental soundscapes (asteroid drones, black hole ambience) became inaudible.

CAUSE: Spatial audio's master gain was unintentionally piped through the UI master gain node (kept at 0.045 to keep click sounds quiet), multiplying environmental audio by the same factor — a ~22x volume reduction.

FIX: Re-routed spatial audio's master gain to connect directly to ctx.destination, bypassing the UI gain stage entirely; also wrapped AudioNode.disconnect() calls in try/catch to survive rapid map-switch teardowns.

The WebSocket CORS Wall

[HIGH] Logic Arena v2.0.0

Badges: WebSocket · CORS · Production

SYMPTOM: Every page using the global socket crashed with 'n.find is not a function' immediately after production deployment.

CAUSE: @WebSocketGateway hardcoded origin: 'http://localhost:3000', silently rejecting all WebSocket upgrades from logicarena.dev before a connection ever formed — the .find() crash was just the visible symptom.

FIX: Extended the CORS origin array to include the production domains and updated Nginx's /socket.io/ location block with proxy_read_timeout 86400 and correct Connection "upgrade" headers for long-lived tunnels.

The Keyboard Blindfold

[HIGH] Logic Arena v3.1.0

Badges: Mobile UX · @dnd-kit · Accessibility

SYMPTOM: The arena requires landscape on mobile, but tapping the code editor made the system keyboard cover 80% of the screen — hiding both code and arena at once.

CAUSE: The mobile keyboard is a system-level overlay; no CSS layout can constrain or coexist with it in landscape mode.

FIX: Eliminated the keyboard as a concept entirely — built a drag-and-drop Block Editor (@dnd-kit) with a recursive block-to-AllScript compiler, so players compose full programs without ever invoking the system keyboard.

The Nginx Route Hijacker

[MEDIUM] Logic Arena v2.0.0

Badges: Nginx · Reverse Proxy · Routing

SYMPTOM: API calls to /campaign/levels, /tournaments, and /auth/login returned HTML instead of JSON after deployment.

CAUSE: Nginx routed every request to the Next.js frontend on port 3000, including backend API paths, because the location blocks were incomplete.

FIX: Added an explicit regex location block routing backend paths to port 3001, with a separate dedicated /socket.io/ block to avoid prefix-matching conflicts with the catch-all frontend route.

The Double /api Prefix

[MEDIUM] Logic Arena v2.1.0

Badges: OAuth · Environment Config

SYMPTOM: GitHub OAuth redirected to /api/api/auth/github/callback, and local dev requests were blocked by CORS from the production CLIENT_URL.

CAUSE: API_BASE_URL was built by appending /api to a BASE_URL that already contained /api; the server's CLIENT_URL pointed at production, blocking all localhost-origin requests during local development.

FIX: Collapsed both URL constants into one env-aware value, and created a fully separate 'Logic Arena Local' GitHub OAuth App with its own callback URL and .env.local loaded via dotenv-cli.

The Admin Rate Limit Wall

[MEDIUM] Logic Arena v3.1.0

Badges: NestJS Throttler · Admin Dashboard

SYMPTOM: Navigating the admin analytics dashboard triggered cascading 429 errors that froze the interface within minutes.

CAUSE: The global rate limiter treated admin traffic identically to public traffic; admin pages fire several parallel API calls on load that collectively exceeded the per-IP threshold.

FIX: Introduced a dedicated admin throttle tier at 300 req/min, added a shared 429 retry helper with backoff across all admin hooks, and staggered simultaneous page-load fetches by 500ms.

The Stasis Shader Freeze

[HIGH] Logic Arena v3.6.x

Badges: WebGL · Shaders · GPU Performance

SYMPTOM: Entering STASIS caused a visible 1-2 second screen freeze as the point-light glow effect popped in.

CAUSE: The StasisEffect and RobotModel point lights were only mounted into the scene graph conditionally, forcing the GPU to compile their shaders on-demand the first time STASIS actually triggered mid-match.

FIX: Kept the point lights unconditionally mounted for every active robot at all times, driving visibility purely through the visible prop and animating intensity (0 when inactive) — shaders now compile once at load time, never mid-combat.

The Camera Sweep Stutter

[MEDIUM] Logic Arena v3.6.x

Badges: WebGL · Three.js · Loading Pipeline

SYMPTOM: Rotating the free-orbit camera around the arena produced visible frame drops the first time it swept past distant scenery.

CAUSE: Off-screen procedural planet, nebula, pulsar, and black hole shaders were compiled lazily on first appearance in the camera frustum instead of upfront.

FIX: Integrated drei's Preload component inside SceneCanvas to force-compile every off-screen shader during the loading screen, so no shader ever compiles mid-gameplay.

The Redis Leaderboard Blackout

[HIGH] Logic Arena v3.6.x

Badges: Redis · Sorted Sets · Data Integrity

SYMPTOM: Registered, active players silently vanished from the leaderboard while still showing correct stats on their own profile.

CAUSE: The Redis sorted set backing the rankings had drifted out of sync with the Postgres user count with no reconciliation check, so missing entries just never rendered.

FIX: Added a size comparison between the Redis sorted set and the database user count on read, triggering a dynamic rebuild of the ranking set from Postgres whenever a drift is detected.

The GHCR Namespace Lock

[MEDIUM] Logic Arena v3.6.x

Badges: CI/CD · GitHub Actions · Docker

SYMPTOM: The deploy pipeline started failing Docker publishes with 'repository name must be lowercase' after transferring the repo to an organization.

CAUSE: Image tags were built from github.repository_owner directly, which can contain mixed case, and the namespace was still hardcoded to the old personal account.

FIX: Added a pipeline step to lowercase repository_owner into an env var for image tags, and replaced the hardcoded personal namespace with the dynamic owner value for seamless publishing under the new org.

The Hidden Contender

[HIGH] Logic Arena v3.6.5

Badges: Socket.io · Guest Auth · Race Condition

SYMPTOM: Guest users entering the arena frequently saw a frozen, empty grid with neither robot rendering.

CAUSE: The socket connection handler checked for a ready script exactly once on connect, but the async guest-script resolver hadn't yet populated the tracking ref — so joinMatch was silently skipped and the server never learned the guest was there.

FIX: Added a joinMatchSentRef state-guard and fired the join emission from three checkpoints — initial connection, successful authentication, and a watcher effect that fires the moment the async guest script resolves.

The Auth Spinner of Doom

[MEDIUM] Logic Arena v3.1.0

Badges: SMTP · Nodemailer · UX

SYMPTOM: Registration and password-reset buttons spun for minutes before timing out, indistinguishable from a broken server.

CAUSE: Gmail's SMTP handshake can silently hang for 2-5 minutes, and nodemailer had zero timeout configured — the server waited indefinitely.

FIX: Added an 8-second connectionTimeout/socketTimeout to nodemailer; on timeout the account is created and the user redirected immediately with an informational banner instead of blocking the whole flow.

The networkidle Stall

[HIGH] Scout v0.24

Badges: Playwright · Wellfound · Performance

SYMPTOM: Every search stalled roughly 30 seconds on login and another ~10 seconds per job before evaluation even began.

CAUSE: login() and hydrateJobDetails() waited on page.waitForLoadState('networkidle'), which never resolved because Wellfound fires continuous background analytics traffic.

FIX: Swapped networkidle for domcontentloaded bounded by a 10-second timeout, restoring near-instant DOM evaluation.

The Socket Eviction Death Loop

[HIGH] Scout v0.24

Badges: Socket.io · WebSocket · Desktop Agent

SYMPTOM: The desktop agent disconnected and re-authenticated every ~6 seconds, spawning a fresh AgentSession row each time.

CAUSE: An unused browser-side socket client connected to the same /agent namespace; the gateway evicts existing sockets on new auth, so the browser tab and desktop agent kept kicking each other off.

FIX: Removed the dead browser-side agent socket client entirely — all dashboard traffic already ran through the separate /dashboard namespace.

The Plaintext JWT Leak

[HIGH] Scout v0.24

Badges: Security · Pino · Logging

SYMPTOM: Full HttpOnly JWT cookies and Authorization headers were printed in plaintext in server logs.

CAUSE: nestjs-pino had no redact configuration, so its default request serializer logged req.headers.cookie / authorization verbatim, including nested error contexts.

FIX: Added explicit redact paths (cookie, authorization, set-cookie — top-level and wildcard-nested) to the Pino config; verified logs now show [REDACTED] globally across all endpoints.

The Sentry Development Freeze

[HIGH] Logic Arena v3.4.0

Badges: Sentry · Webpack · Next.js

SYMPTOM: Running Next.js dev server froze the entire machine within seconds — mouse unresponsive, hard restart required.

CAUSE: widenClientFileUpload: true told webpack to scan the whole project for source-map manifests; combined with Session Replay SDK spawned CPU-saturating worker threads.

FIX: Removed Sentry entirely — uninstalled @sentry/nextjs, deleted all config files, restored next.config.ts to export nextConfig directly.

The Poisoned Redis Flag

[HIGH] Logic Arena v3.4.0

Badges: Redis · ioredis · State Management

SYMPTOM: Users requesting password reset received no email and no error — Redis write silently dropped.

CAUSE: isReady set to false inside the error handler, which fires on transient network hiccups that the client auto-recovers from.

FIX: Removed the false assignment from the error handler; isReady only flips on reconnecting and end events.

The Blocked Mail Server

[HIGH] Logic Arena v3.4.0

Badges: SMTP · Resend · DigitalOcean

SYMPTOM: All transactional emails silently dropped since production launch with no error in logs.

CAUSE: DigitalOcean blocks outbound SMTP port 465 on all Droplets; nodemailer connection timeout looked like a delivery queue delay.

FIX: Replaced nodemailer with Resend HTTP SDK (bypasses SMTP entirely), verified SPF/DKIM/DMARC.

Redis IPv6 Resolution in Docker

[HIGH] Logic Arena v2.0.0

Badges: Redis · Docker · Networking

SYMPTOM: Upstash Redis connection collapsed in Docker with getaddrinfo ENOTFOUND, looking like auth failure.

CAUSE: ioredis silently resolved hostname to an IPv6 address unreachable inside the container's network stack.

FIX: Forced IPv4 with family: 4 in ioredis config and switched to explicit TLS handshake on port 6379.

The Prisma Ghost Engine

[HIGH] Logic Arena v2.0.0

Badges: Prisma · Docker · Alpine

SYMPTOM: Container booted, mapped all routes, connected to Redis, then crashed: 'Unable to require libquery_engine-linux-musl.so.node'.

CAUSE: Prisma client generated on Windows for wrong binary target; Alpine container lacked openssl.

FIX: Added linux-musl targets to schema.prisma, installed openssl via apk, copied .prisma client from builder to runner stage.

Operator Precedence Disaster

[HIGH] Logic Arena v2.5.0

Badges: Compiler · Parser · AST

SYMPTOM: $2 + 3 * 4$ evaluated to 20 instead of 14 — all AliScript math with mixed operators was silently wrong.

CAUSE: AliScript expression parser evaluated all binary operators left-to-right with equal precedence.

FIX: Split parseBinaryExpression into parseAddition and parseMultiply establishing proper precedence tower.

The FOV Fire Hack

[HIGH] Logic Arena v2.5.0

Badges: Game Engine · Combat · FOV

SYMPTOM: Robot with scanner pointing East could FIRE at enemy standing directly behind it — the vision cone was cosmetic.

CAUSE: FIRE/BURST_FIRE selected targets by searching all living robots, completely ignoring the FOV cone.

FIX: Changed target selection to use robot.visibleEntities.robots exclusively; if cone is empty, FIRE returns with zero energy cost.

WebGL Context Loss (60Hz console.log)

[HIGH] Logic Arena v3.5.0

Badges: WebGL · R3F · GC Pressure

SYMPTOM: Arena tab froze entirely — full browser hang requiring a tab kill — seemingly at random.

CAUSE: console.log inside the 60Hz useFrame loop serialized and heap-allocated the delta object 60 times/sec, triggering GC and WebGL context reclaim.

FIX: Removed the console.log from the GameState delta processor entirely.

Texture Destruction Freeze

[HIGH] Logic Arena v3.6.0

Badges: Three.js · useGLTF · WebGL

SYMPTOM: Reconnecting to a match crashed the WebGL canvas or rendered robot models completely invisible.

CAUSE: Unmount cleanup called .dispose() on materials/textures of shared GLB assets that useGLTF caches globally, corrupting the shared cache.

FIX: Removed all manual material/texture dispose calls; delegated teardown to a React key driven by the cloned scene UUID.

Broken Docker Container Assembly

[HIGH] Logic Arena v3.6.0

Badges: Docker · TypeScript · Monorepo

SYMPTOM: Production Docker crashed with 'Cannot find module @logic-arena/engine/constants' — local builds passed fine.

CAUSE: Local pnpm symlinks resolved workspace subpaths that the clean Docker layer lacked; Dockerfile built server before shared packages.

FIX: Built logic-parser and engine sequentially before server stage, added wildcard tsconfig subpath aliases, copied root tsconfig.json into build layer.

Deterministic TLE Quota System

[HIGH] Logic Arena v2.7.0

Badges: Sandboxing · AST · Fairness

SYMPTOM: Same AliScript passed on a gaming PC but failed on a low-end laptop — infinite-loop limit was hardware-dependent.

CAUSE: MAX_TICK_DURATION_MS measured wall-clock milliseconds, making execution limit non-deterministic.

FIX: Replaced timestamp gate with platform-agnostic MAX_OPERATIONS_PER_TICK quota (2,000 ops) tracked via opsCounter reference through block executor.

The Anti-Corruption Layer

[HIGH] Cybership API

Badges: DDD · TypeScript · Zod

SYMPTOM: Runtime crashes whenever external carrier returned unexpected JSON shape that bled into core business logic.

CAUSE: Unvalidated third-party API structures trusted directly inside the domain layer.

FIX: Implemented Anti-Corruption Layer using Zod for strict runtime schema enforcement at every external boundary.

WebRTC Signaling Dilemma

[HIGH] Flurry v2.0

Badges: Socket.io · WebRTC · P2P

SYMPTOM: P2P audio/video calls suffered unpredictable connection states and noticeable latency.

CAUSE: Complex WebRTC handshakes and ICE candidate exchange had no reliable orchestration layer.

FIX: Architected hybrid signaling server on Socket.io to manage handshakes while offloading media to WebRTC P2P channels.

The GraphQL That Lied With Zeros

[HIGH] Portfolio v2.0.0

Badges: GraphQL · Next.js · API Safety

SYMPTOM: GitHub Activity card showed 0 for Commits, Contributions, and Day Streak while other stats showed correctly.

CAUSE: Revoked GitHub PAT caused GraphQL to return 401; loose optional chaining silently resolved to undefined — zero console logs.

FIX: Rotated token, added strict .ok status checks on REST/GraphQL, checked gql.errors in response body with immediate console logging.

The REPLAN Infinite Loop

[MEDIUM] Scout v0.21

Badges: LLM Reasoning · Agent Loop · State Machine

SYMPTOM: The agent looped forever re-navigating to the jobs page, wiping out its own pending apply-filter and discover-jobs tasks.

CAUSE: The Reasoner re-evaluated every cycle; a stale DOM snapshot during navigation made the LLM recommend REPLAN, which unconditionally erased the plan queue.

FIX: Bypassed REPLAN when the plan queue already has valid pending tasks and the previous task executed cleanly.

The Reserved Port Redis Block

[MEDIUM] Scout v0.20

Badges: Windows · WSL2 · Redis

SYMPTOM: The API refused to boot with ECONNREFUSED on Redis, and the desktop agent looped on xhr poll error.

CAUSE: Windows Hyper-V/WSL2 NAT dynamically reserved port range 6173–6672 on boot, blocking host socket binding on the default Redis port.

FIX: Remapped Redis and Qdrant to host ports outside the reserved range across docker-compose.yml and all .env files.

Docker Context Bomb (2.57 GB)

[MEDIUM] Logic Arena v2.0.0

Badges: Docker · pnpm · Monorepo

SYMPTOM: docker compose up --build transferred 2.57 GB to Docker daemon and hung the machine for 10 minutes.

CAUSE: .dockerignore missing node_modules exclusion — full workspace dependency tree sent as build context.

FIX: Rewrote .dockerignore with **/node_modules globs, switched pnpm to --shamefully-hoist to flatten binaries.

Circular Jitter Navigation Loop

[MEDIUM] Logic Arena v1.8.0

Badges: A* · Pathfinding · Weighted Grid

SYMPTOM: Robots entered spasmodic oscillation near traps, spinning in place instead of moving.

CAUSE: Binary pass/block A* combined with waypoints landing on the robot's own cell created a recursive 'pointing-at-self' feedback loop.

FIX: Overhauled heuristic to Weighted Cost Grid (TRAP 3.0, LAVA 5.0) and added Self-Waypoint Skip consuming waypoints within half-cell radius.

Cascading Filter Race Condition

[MEDIUM] CS Arena

Badges: Next.js 15 · Server Components · useTransition

SYMPTOM: Clearing several dependent filters at once caused UI jank and router race conditions.

CAUSE: Heavy URL state-sync ran on the main UI thread, blocking navigation when multiple dependent states changed together.

FIX: Used Next.js 15 useTransition and useActionState to decouple URL state-sync from the UI thread.

Offline-First Illusion

[MEDIUM] Flurry v2.0

Badges: PWA · Inngest · Service Workers

SYMPTOM: App appeared to freeze or lose data during intermittent connectivity, especially mid multi-step message processing.

CAUSE: User actions had no durable queue — anything in-flight when connection dropped was lost.

FIX: Engineered sync layer with Inngest durable background functions, queuing actions in IndexedDB and re-hydrating via Workbox on reconnection.

// WORK EXPERIENCE

Frontend Lead | South Valley National University — Field Training

Feb 2026 – Jun 2026

- Led frontend of StudentHub (campus platform) in an 8-person Agile team: 500+ commits, 135+ PRs.
- Built 47 web screens in React 18 + Vite + Tailwind CSS with full EN/AR bilingual RTL localisation.
- Integrated JWT-based role authorisation and connected to a .NET Web API with 110+ REST endpoints.
- Designed Campus Store module and 6 core student service modules.

Backend Intern | Web Masters — Remote, Cairo

Jul 2025 – Aug 2025

- Engineered RESTful APIs for a CMS (Blog Pro) and a URL Shortener using Node.js / Express / MongoDB.
- Integrated Firebase Realtime Database for instant messaging — reduced data sync latency by ~40%.
- Optimised MongoDB schemas with Joi validation — accelerated query execution by ~30%.
- Implemented JWT authentication ensuring 100% secure end-to-end data access.

// EDUCATION

B.Sc. Computer Science & Artificial Intelligence

South Valley National University, Qena, Egypt | Oct 2023 – Jun 2027

Relevant coursework: Data Structures & Algorithms, OOP, Database Systems, Software Engineering, Operating Systems.

// ESTABLISHED WORKFLOW RULES (always apply when working with Ali)

- Shell syntax: PowerShell — use semicolons (;), never &&.
- CSS variables only — zero hardcoded hex or Tailwind palette color classes.
- No TypeScript any — ever.
- type='button' on all buttons; NOT on anchor tags used for navigation.
- 44px minimum touch targets on all interactive elements.
- Framer Motion must be disabled on mobile via useIsMobile hook; value computed once in parent, passed as prop.
- Package manager: pnpm for Logic Arena; npm for Portfolio; pnpm + Turborepo for the Scout monorepo.
- All prompts to coding agents (Cline / GitLab Duo) must be written in English only.
- Tokens and credentials must never be shared in chat — not even partial values.
- AI coding agent (Cline or GitLab Duo Chat) handles all implementation; Ali uses Claude for prompt engineering and plan review only.

// CONTEXT NOTE FOR AI ASSISTANTS

This document is Ali's living technical profile. When Ali shares this with you he wants you to understand his current project state, stack, and real engineering experience without him having to re-explain from scratch. Key points:

- Logic Arena is his primary flagship — treat all its docs as authoritative context.
- He does NOT want you to implement code — he uses Cline/GitLab Duo for that. Your job is planning, reviewing, and prompt-engineering.
- Respect the workflow rules above in every suggestion.
- His portfolio is at aliaggag.me — the codebase uses Next.js with a strict CSS variable design system.
- Arshivi is a side project: vanilla stack + Supabase + Cloudflare Pages.
- Scout is his current flagship AI agent project (graduation project, dominant focus since mid-July 2026) — treat its architecture docs (Agent Constitution, V2 Migration Blueprint) as authoritative, and note the ethical constraints (Final Click Rule, no anti-bot evasion) are permanent and non-negotiable.

// CERTIFICATIONS

- Front-End Development Diploma (React.js) — Sef Academy
- Back-End Development Training Program — Web Masters
- Back-End Development Diploma (Node.js) — Sef Academy
- The Web Frontend Learning Guide — Udemy